Game 4 Animal Quiz

Before you start

These are teacher-designed Scratch 3 instructions. Work through the core game first, then test it with a partner. Save your project in the class folder.

Ask one question

Choose an animal sprite and a simple backdrop. Begin with one question and a typed answer. Build this script on the sprite:

```
when green flag clicked
ask [Which animal says meow?] and wait
if <(answer) = [cat]> then
  say [Correct!] for (2) seconds
else
  say [A cat says meow. Try the next one.] for (2) seconds
end
```

The answer reporter comes from Sensing. The equals operator compares the answer with the expected word.

Add two more questions

Duplicate the ask-and-check section below the first question, inside the same green-flag script. Change both the question and the expected answer.

Try: Which animal says woof? Expected answer: dog. Which animal has a trunk? Expected answer: elephant.

Keep each question short. Agree acceptable words in advance; the first version accepts the specified word, not every possible sentence.

Test both branches

Enter the correct answer for each question, then restart and deliberately enter an incorrect answer. Check that feedback matches the question.

If an answer is wrongly rejected, check spelling and accidental spaces in the expected answer. If the wrong feedback appears, check the duplicated question's answer block.

Personalise and explain

Change the animal, backdrop or questions after the first quiz works. Keep the first quiz to two or three questions.

Show the block that asks a question, the answer reporter and the two feedback paths.

Sources and related skills

Related book pages are included after this guide. They teach supporting skills; this game is a separate classroom project.

Official Scratch project ideas: https://scratch.mit.edu/ideas

3.3 An algorithm is a plan

In this lesson

You will learn:

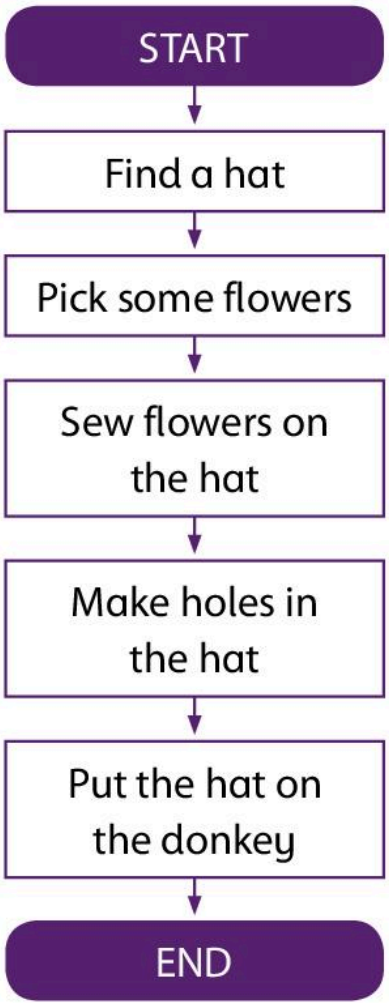
➔ what an algorithm is.

A plan that tells you how to do a task is called an **algorithm**.

The algorithm must include all the actions. The actions must be in the right order.

You can draw an algorithm. Put the actions in boxes. Join the boxes with arrows to show the right order.

Here is an algorithm for making a donkey's hat.



Activity



A big brother made a toy horse for his little sister. Here are the actions.

Find some wood



Carve the wood into a horse shape



Let the paint dry

Paint the horse



Buy some paint

Write the actions in the right order.

Extra challenge

Make an algorithm for making a toy horse. Write the actions in boxes joined with arrows. On a big piece of paper make a colourful picture.

Think again

What does 'algorithm' mean? Write or say the answer in your own words.

3.4 Algorithms and programs

In this lesson

You will learn:

- how making an algorithm helps you to plan a program.



Program

A **program** is a series of commands. The commands tell the computer what to do. When you **run a program** the computer carries out all the commands in the program.

A **programmer** is the person who makes a computer program.

A programmer sometimes makes an algorithm before they make the program. The algorithm is the plan for the program.

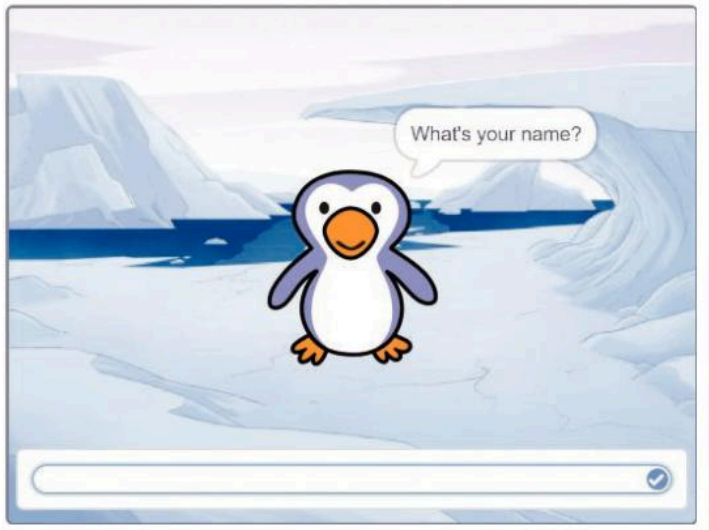
Sprite

You can make a program. You use a programming language to make programs. Scratch is a programming language.

- A Scratch program controls a **sprite**. A sprite is an object or creature on the screen. The program makes it do different things.
- A Scratch program is also called a **script**. A script is a short program.

Activity

A programmer wanted to make a program. The sprite is a penguin. Here is a list of things that the penguin will do.



Say goodbye

Go off the screen

Say hello and your name

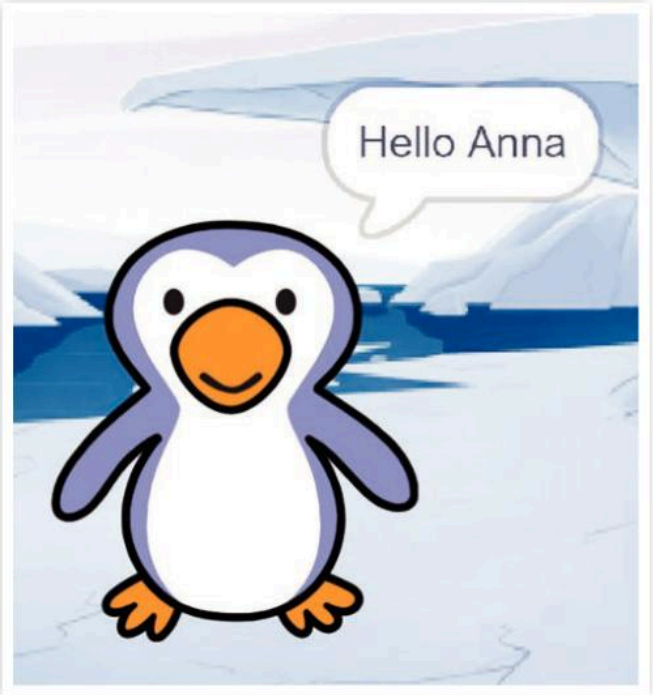
Ask your name

Come onto the screen

Think again

Write an algorithm showing the actions of the program in boxes joined by arrows.

Put these actions in the right order.



Extra challenge

Think of one extra action the penguin could do. Make a plan for the program that includes the extra action.

Be creative

Plan a new computer program:

- Draw the sprite.
- Say what actions the sprite will do.

Check what you know

You have learned

- how to plan a program
- what an algorithm is
- what happens when you run a program.



Activity

Here is a plan for the actions in a quiz program. They are not sorted into order.



Tell you if the answer is right

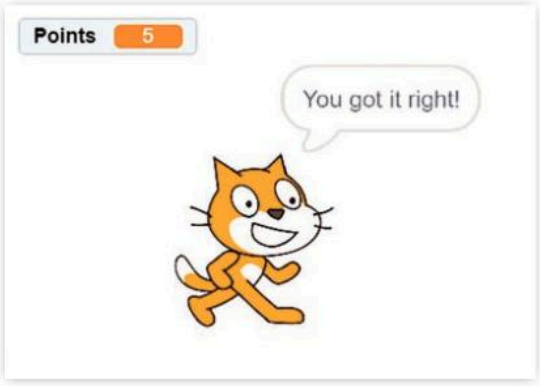
Show how many points you have got

Ask a question

Give you a point for a right answer

Get the answer to the question

Make an algorithm by putting the actions in the right order.



Test

A programmer made a program. Here are the three things she did.

Plan the program
Write the program
Run the program

1. Write the three actions in your book.
2. Tick the action that means carry out the commands.
3. Put a star next to the one that means make an algorithm.
4. Fill in the missing words in these sentences. Use the words from the box.

actions	algorithm	sequence	program

An _____ is a plan to make a _____.

It sets out all the _____ in the right _____.

Self-evaluation

- I answered test questions 1 and 2.
- I answered test questions 1–3.
- I did some of the activity.
- I answered all the test questions.
- I did all of the activity.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?